Scene Assessment & Data Source Identification

Pre-Analysis: Open Source Intelligence (OSINT)

Before setting foot on scene — or connecting to a network — conduct thorough **pre-analysis** using open source intelligence gathering techniques. This preparatory phase is critical for understanding the scope and nature of the environment you are about to examine.

Domain & IP Research

Query WHOIS records, DNS histories, and IP geolocation data to understand the digital footprint of the subject or organisation under investigation.

Social & Public Records

Review publicly available social media profiles, corporate filings, and breach databases for contextual intelligence on potential accounts and services in use.

Technology Stack Profiling

Identify likely operating systems, cloud providers, and software platforms through job postings, vendor disclosures, or technology fingerprinting tools.

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Initial Survey of the Scene

Whether approaching a **physical location** or a **network environment**, the initial survey establishes the boundaries of the investigation and identifies immediate priorities. This step must be systematic and documented from the outset.

Physical Scenes

Photograph the environment, note device states (powered on/off), record cable connections, and document the layout of workstations and server rooms.

Network Environments

Capture running processes, active connections, logged-in users, and volatile memory states before any changes occur. Time is critical.

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Identify Key Devices & Data Locations

A thorough identification of all devices and storage locations is the foundation of a sound forensic strategy. Overlooking even a single data source can compromise the integrity of the entire investigation.

Local Storage

Internal hard drives, SSDs, and embedded storage within desktops, laptops, and mobile devices. Check BIOS/UEFI for hidden or secondary drives.

Remote Servers

On-premises file servers, database servers, mail servers, and backup systems. Verify access permissions and retention policies.

Cloud Services

SaaS platforms, IaaS instances, cloud storage (e.g. OneDrive, Google Drive, AWS S3). Require legal authorisation and provider cooperation for acquisition.

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Check for Connected & Peripheral Devices

Investigators must extend their search beyond the obvious. Connected peripherals and ancillary devices frequently hold **overlooked but highly relevant evidence** — from print job logs to removable media containing exfiltrated data.

Printers & Scanners

Modern printers retain job logs, cached documents, and user authentication records on internal storage.

Removable Media

USB drives, SD cards, external HDDs, and optical media. Check for recently mounted devices in OS logs.

Network-Attached Devices

NAS units, IoT devices, IP cameras, and smart home hubs may contain logs, footage, or synced data.

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Map All Potential Data Sources

Creating a comprehensive map of all data sources ensures nothing is missed during acquisition. Use **network topology diagrams** and **asset inventories** to build a complete picture of the digital environment.

Network Topology Diagrams

- Document all network segments, VLANs, and subnets
- Identify gateways, firewalls, and proxy servers
- Map traffic flows between internal and external systems
- Note wireless access points and guest networks

Asset Inventories

- Cross-reference with IT asset management databases
- Verify serial numbers, MAC addresses, and hostnames
- Identify shadow IT — unapproved devices and services
- Document software licences and installed applications

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Ephemeral Data: The Hidden Evidence

Perhaps the most frequently overlooked — and most volatile — category of evidence resides in **ephemeral storage**. This data is transient by nature and may be lost within seconds if not identified and preserved immediately.

Cloud Syncing

Files synced to services like Dropbox, iCloud, or Google Drive may differ from local copies. Sync logs, version histories, and deletion records provide critical timeline evidence that disappears with account changes.

Hidden Sectors & HPA

Host Protected Areas (HPA) and Device Configuration Overlays (DCO) on hard drives can conceal data invisible to standard tools. Specialised forensic utilities are required to detect and image these regions.

Volatile Memory (RAM)

Running processes, encryption keys, clipboard contents, and active network connections exist only in RAM. Powering down a device destroys this evidence irreversibly — capture it first.

Messaging & Temp Files

Ephemeral messaging apps (Signal, Snapchat), browser sessions, temporary caches, and pagefile data may contain crucial artefacts with short retention windows.

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Putting It All Together: The Assessment Workflow

Pre-Analysis **Initial Survey** **Identify & Catalogue** **Map Data Sources**

Each phase builds upon the previous. Skipping or rushing any step risks compromising evidence integrity and creating gaps that opposing counsel or defence teams will exploit. Thoroughness at this stage pays dividends throughout the entire investigation lifecycle.

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Key Takeaways

1. **Always conduct OSINT before arriving on scene**

   Pre-analysis reduces surprises and enables you to prepare the correct tools and legal authorities in advance.

2. **Survey systematically, document relentlessly**

   Every observation, device state, and connection must be recorded. Photographs, notes, and chain-of-custody logs are non-negotiable.

3. **Look beyond the obvious devices**

   Peripherals, network-attached storage, cloud services, and shadow IT frequently harbour the most probative evidence.

4. **Never underestimate ephemeral data**

   Volatile memory, cloud sync artefacts, hidden disk sectors, and temporary files can make or break your case — but only if captured in time.

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Thorough Preparation. Sound Investigation.

The quality of your scene assessment directly determines the quality of your forensic conclusions. **Invest the time upfront** — every device catalogued, every data source mapped, and every ephemeral artefact preserved strengthens the evidentiary foundation of your case.

Remember: you cannot acquire what you have not identified. A comprehensive scene assessment is not merely best practice — it is a professional obligation.

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